

Gopi M

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SUMMARY

Final-year Data Science student specializing in LLM systems and RAG architectures, with hands-on experience building scalable AI pipelines, fine-tuning transformer models, and deploying production-ready ML systems. Strong foundation in Python, PyTorch, and end-to-end model deployment. Seeking AI/ML internship roles.

SKILLS

Languages:	Python, SQL, R
AI/ML:	LLM Fine-tuning (QLoRA, PEFT), RAG Systems, ML Pipeline Development
Frameworks:	PyTorch, Scikit-learn, HuggingFace Transformers
Backend:	FastAPI, REST APIs, Async Programming
Tools:	LangChain, FAISS, ChromaDB, Pandas, NumPy, Matplotlib
Cloud/DevOps:	Docker, GCP (Cloud Run, Vertex AI), CI/CD, Git
Databases:	PostgreSQL, MySQL, Supabase

EXPERIENCE

Machine Learning Engineer Intern <i>Resilience Business Grids LLP</i>	Nov 2024 – Jun 2025 <i>Coimbatore, India</i>
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- Engineered and deployed computer vision and deep learning models using PyTorch, covering the full ML lifecycle including preprocessing, training, evaluation, and inference via REST APIs.
- Optimized model performance through hyperparameter tuning, learning rate scheduling, and architecture improvements, increasing model efficiency and stability across CNN and transformer-based systems.
- Integrated ML models into production web applications using Django and React, enabling scalable real-time inference workflows.

PROJECTS

Navigator – AI Desktop Automation Assistant | Electron, React, Node.js, TypeScript

- Architected an LLM-based desktop automation system capable of executing browser tasks, file operations, and document workflows, reducing manual effort in repetitive tasks.
- Orchestrated multiple MCP servers to enable modular agent execution and cross-platform automation with Telegram integration.
- Designed a permission-based system with encrypted API key storage and a real-time task execution interface.

Toxicity Reduction in Falcon 7B using RLAIIF | PyTorch, QLoRA, PPO

- Built a model alignment pipeline to reduce toxic outputs in large language models for legal-domain applications.
- Implemented Constitutional AI and Reinforcement Learning from AI Feedback (RLAIF) using PPO to improve response safety.
- Fine-tuned Falcon 7B on domain-specific datasets, achieving a 75% reduction in toxicity while retaining 92% model accuracy.

EDUCATION

Integrated MSc Data Science <i>Amrita Vishwa Vidyapeetham</i>	2022 – 2027 <i>Coimbatore</i>
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- Coursework: Machine Learning, Deep Learning, DBMS, Computer Networks

PUBLICATIONS & RESEARCH

VG-RAG: Verification-Gated Retrieval-Augmented Generation

Jan 2026 – Present

- Proposed a retrieve-verify-generate architecture to improve reliability in RAG-based systems.
- Designed a multi-stage verification mechanism combining semantic relevance, consistency checks, and cross-document validation.
- Implemented conditional web verification to filter unreliable retrievals and improve factual accuracy.

ACHIEVEMENTS

1st Place – Big-O-Battle Coding Competition | PSG College of Technology

2025

- Engineered an autonomous game-playing bot that ranked **1st among 91 teams** in a competitive round-robin tournament using optimized algorithms and efficient strategy design.

3rd Place – Code O’Clock National Hackathon | SaaS22 × CIT

2025

- Designed *Mentora*, an AI-driven course generator that converts PDFs, text, and video into structured learning paths with multilingual support and automated quizzes, delivered within **24 hours**.

OPEN SOURCE & CONTRIBUTIONS

English–Tamil Neural Machine Translation Model (M2M100, 424M)

2025

- Fine-tuned a transformer-based translation model using custom preprocessing, BPE tokenization, and training pipelines for English–Tamil language tasks.
- Published on [HuggingFace Hub](#), achieving **100+ downloads** and supporting Tamil NLP research and downstream applications.

Gitsy – AI Git Workflow Assistant (VS Code Extension)

2025

- Developed a VS Code extension integrating AI-based pre-commit validation, including secret detection, branch safety checks, and code quality analysis.
- Available on [VS Code Marketplace](#), featuring an interactive dashboard for real-time repository insights and Git workflow management.